Lecture 27

Invariant tensors and commutants

Objectives. *We identify intertwiners with invariant tensors. This gives commutants from multiplicity spaces and explains classical contractions. We also determine the representation ring.*

By the end of this lecture you will be able to:

- Translate between intertwiners, invariant tensors and invariant bilinear forms.
- Compute a commutant from multiplicity spaces.
- Distinguish O_N-invariants from $\mathfrak{so}_N$-invariants, including volume tensors.
- Prove that the representation ring is a polynomial ring in the fundamental characters.

Throughout, $\mathfrak{g}$ is a nonzero complex semisimple Lie algebra, and modules are finite-dimensional. The named classical examples also use $\mathfrak{gl}_n$, the zero algebra $\mathfrak{sl}_1$, the abelian algebra $\mathfrak{so}_2$ and matrix groups. Recall the dual and tensor actions from Lecture 14:

$$(x\xi)(u) = -\xi(xu), \qquad x(u \otimes w) = xu \otimes w + u \otimes xw.$$

The *invariants* of a module W are

$$W^{\mathfrak{g}} = \{w \in W : xw = 0 \text{ for all } x \in \mathfrak{g}\} \cong \mathrm{Hom}_{\mathfrak{g}}(\mathbb{C}, W).$$

27.1 Intertwiners as invariant tensors

Proposition 27.1 (Intertwiners and invariants)**.** *Let U and V be modules. The linear isomorphism*

$$\Phi\colon U^* \otimes V \longrightarrow \mathrm{Hom}(U,V), \qquad \Phi(\xi \otimes v)(u) = \xi(u)\, v,$$

satisfies

$$\Phi(xt) = \rho_V(x)\Phi(t) - \Phi(t)\rho_U(x) \quad \text{for all } x \in \mathfrak{g},\ t \in U^* \otimes V.$$

Hence $\mathrm{Hom}_{\mathfrak{g}}(U,V) \cong (U^* \otimes V)^{\mathfrak{g}}$.

Proof. The map Φ is bijective because U and V are finite-dimensional. Both sides are linear in t, so it suffices to take $t = \xi \otimes v$. For $u \in U$, the tensor action, the dual action and the definition of Φ give

$$\begin{aligned} \Phi\big(x(\xi \otimes v)\big)(u) &= \Phi(x\xi \otimes v + \xi \otimes xv)(u) \\ &= -\xi(xu)\, v + \xi(u)\, xv \\ &= \big(\rho_V(x)\Phi(\xi \otimes v) - \Phi(\xi \otimes v)\rho_U(x)\big)(u). \end{aligned}$$

So $xt = 0$ for all x exactly when $\Phi(t)$ intertwines the actions. ∎

For $U = V$, the map Φ turns the evaluation $\xi \otimes v \mapsto \xi(v)$ into the trace:

$$\xi(v) = \operatorname{tr} \Phi(\xi \otimes v).$$

Let (v_j) be a basis of V with dual basis (v_j^*). The invariant tensor $\sum_j v_j^* \otimes v_j$ corresponds to id_V, and its evaluation is $\dim V$.

Corollary 27.2 (Tensor–Hom reciprocity)**.** *For modules U, V, W,*

$$\operatorname{Hom}_{\mathfrak{g}}(U \otimes W, V) \cong \operatorname{Hom}_{\mathfrak{g}}(U, W^* \otimes V).$$

Proof. The linear isomorphism $U^* \otimes W^* \to (U \otimes W)^*$, $(\xi \otimes \eta)(u \otimes w) = \xi(u)\eta(w)$, is equivariant:

$$(x\xi \otimes \eta + \xi \otimes x\eta)(u \otimes w) = -\xi(xu)\eta(w) - \xi(u)\eta(xw) = -(\xi \otimes \eta)\big(x(u \otimes w)\big).$$

Hence this isomorphism and Proposition 27.1, applied twice, give

$$\begin{aligned} \operatorname{Hom}_{\mathfrak{g}}(U \otimes W, V) &\cong \big((U \otimes W)^* \otimes V\big)^{\mathfrak{g}} \\ &\cong \big(U^* \otimes (W^* \otimes V)\big)^{\mathfrak{g}} \\ &\cong \operatorname{Hom}_{\mathfrak{g}}(U, W^* \otimes V). \end{aligned}$$ ∎

A bilinear form B on V is *invariant* if

$$B(xu, v) + B(u, xv) = 0 \quad \text{for all } x \in \mathfrak{g},\ u, v \in V.$$

Equivalently, the linear map $V \otimes V \to \mathbb{C}$, $u \otimes v \mapsto B(u, v)$, is an intertwiner to the trivial module $\mathbb{C}$.

Corollary 27.3 (Invariant bilinear forms)**.** *Let L be irreducible. The invariant bilinear forms on L form a space of dimension one if $L \cong L^*$, and zero otherwise. A nonzero invariant form is nondegenerate and is either symmetric or alternating.*

Proof. Reciprocity with $U = W = L$ and $V = \mathbb{C}$ gives

$$\operatorname{Hom}_{\mathfrak{g}}(L \otimes L, \mathbb{C}) \cong \operatorname{Hom}_{\mathfrak{g}}(L, L^*), \qquad B \longmapsto \big(u \mapsto B(u, \cdot)\big).$$

Schur's lemma gives the dimension of $\operatorname{Hom}_{\mathfrak{g}}(L, L^*)$. A nonzero form corresponds to an isomorphism $L \to L^*$, so it is nondegenerate.

The transpose $B^t(u, v) = B(v, u)$ is again invariant. If $L \cong L^*$, the involution $B \mapsto B^t$ acts on the one-dimensional space of invariant forms by $+1$ or -1. So a nonzero invariant form is symmetric or skew-symmetric, and over $\mathbb{C}$ a skew-symmetric form is alternating. ∎

27.2 Commutants from multiplicity spaces

By Lecture 14, a module V has multiplicity spaces $M_\lambda = \operatorname{Hom}_{\mathfrak{g}}(L(\lambda), V)$, of dimension m_λ, and the evaluation

$$\mathrm{ev}\colon \bigoplus_{\lambda \in P_+} M_\lambda \otimes L(\lambda) \longrightarrow V, \qquad \varphi \otimes u \longmapsto \varphi(u),$$

is an isomorphism. Here $\mathfrak{g}$ acts only on the factors $L(\lambda)$.

Theorem 27.4 (Isotypic commutant formula). *The map*

$$\Psi\colon\ \mathrm{End}_{\mathfrak{g}}(V) \longrightarrow \bigoplus_{\lambda} \mathrm{End}(M_\lambda), \qquad \Psi(T)_\lambda(\varphi) = T \circ \varphi,$$

is an algebra isomorphism. Hence

$$\dim \mathrm{End}_{\mathfrak{g}}(V) = \dim (V^* \otimes V)^{\mathfrak{g}} = \sum_{\lambda} m_\lambda^2.$$

Proof. If φ intertwines, so does $T \circ \varphi$. Thus $\Psi(T)_\lambda \in \mathrm{End}(M_\lambda)$. The map Ψ is linear and respects composition and identities, because $(ST) \circ \varphi = S \circ (T \circ \varphi)$.

Injectivity. If $T \circ \varphi = 0$ for all $\varphi \in M_\lambda$ and all λ, then T vanishes on every image $\varphi(L(\lambda))$. These images span V because ev is onto. Hence $T = 0$.

Surjectivity. Let $A_\lambda \in \mathrm{End}(M_\lambda)$. The operator $\bigoplus_\lambda A_\lambda \otimes \mathrm{id}$ commutes with $\mathfrak{g}$, since $\mathfrak{g}$ acts only on the second factors. Put

$$T = \mathrm{ev} \circ \Big(\bigoplus_{\lambda} A_\lambda \otimes \mathrm{id} \Big) \circ \mathrm{ev}^{-1} \in \mathrm{End}_{\mathfrak{g}}(V).$$

For $\varphi \in M_\lambda$ and $u \in L(\lambda)$,

$$T(\varphi(u)) = T\big(\mathrm{ev}(\varphi \otimes u)\big) = \mathrm{ev}(A_\lambda \varphi \otimes u) = (A_\lambda \varphi)(u).$$

So $T \circ \varphi = A_\lambda \varphi$, and $\Psi(T) = (A_\lambda)_\lambda$. The dimension formula follows, since $\dim \mathrm{End}(M_\lambda) = m_\lambda^2$. Proposition 27.1 gives the middle term. ∎

Hence the commutant $\mathrm{End}_{\mathfrak{g}}(V)$ is commutative exactly when every $m_\lambda \le 1$, that is, when V is *multiplicity-free*. The invariants of V form the multiplicity space of the trivial module: $V^{\mathfrak{g}} \cong M_0$.

For $V = \mathbb{C}^n$ and $\sigma \in S_d$, the *permutation operator* is

$$\sigma(v_1 \otimes \cdots \otimes v_d) = v_{\sigma^{-1}(1)} \otimes \cdots \otimes v_{\sigma^{-1}(d)}.$$

Corollary 27.5 (Commutant in type A). *Let $V = \mathbb{C}^n$ and $d \ge 1$. The commutants of $\mathfrak{gl}_n$ and of $\mathfrak{sl}_n$ on $V^{\otimes d}$ both equal the span of the permutation operators. Moreover,*

$$\big(V^{\otimes k} \otimes V^{*\otimes l}\big)^{\mathfrak{gl}_n} = 0 \quad \text{for } k \neq l.$$

Proof. For $n = 1$, every tensor space is one-dimensional. Permutations act as the identity, so their span is the commutant of either algebra. The mixed-tensor assertion follows since I acts by $k - l$. Assume $n \ge 2$. Since $\mathfrak{gl}_n = \mathfrak{sl}_n \oplus \mathbb{C} I$ and the identity matrix I acts on $V^{\otimes d}$ by d id, the two commutants agree. The finite-group course gives the Schur–Weyl decomposition

$$V^{\otimes d} \cong \bigoplus_{\lambda \vdash d,\ \ell(\lambda) \le n} \mathbb{S}_\lambda(V) \otimes V^\lambda.$$

Here V^λ is the Specht module of S_d, and S_d acts only on these factors.

By Lecture 18, the $\mathbb{S}_\lambda(V)$ are irreducible $\mathfrak{sl}_n$-modules, and two of them have the same highest weight only if the partitions differ by full columns of length n. Two different partitions of d cannot differ in this way, so the $\mathbb{S}_\lambda(V)$ are pairwise nonisomorphic. Their multiplicity spaces, indexed here by partitions, are $M_\lambda \cong V^\lambda$. The permutation σ acts on M_λ through its action on V^λ.

Theorem 27.4 gives

$$\operatorname{End}_{\mathfrak{gl}_n}(V^{\otimes d}) \cong \bigoplus_{\ell(\lambda)\le n} \operatorname{End}(V^\lambda).$$

By Wedderburn's theorem for $\mathbb{C}[S_d]$, the map $\mathbb{C}[S_d] \to \bigoplus_{\lambda\vdash d} \operatorname{End}(V^\lambda)$ is onto. So is its projection onto the summands with $\ell(\lambda) \le n$. Hence the permutation operators span $\operatorname{End}_{\mathfrak{gl}_n}(V^{\otimes d})$.

Finally, the identity matrix acts on $V^{\otimes k} \otimes V^{*\otimes l}$ by $(k-l)$ id. An invariant vector is therefore zero when $k \neq l$. ∎

For $k = l$, Proposition 27.1 turns σ into an invariant tensor that pairs the copies of V^* with the copies of V according to σ. For $k = 1$ this tensor is $\sum_i e_i^* \otimes e_i$, the tensor of id_V from Section 27.1.

27.3 Contractions, forms and volume tensors

Let B be a nondegenerate invariant bilinear form on V.

Lemma 27.6 (The invariant tensor of a form). *Let (v_j) be a basis of V, and choose v_j' with $B(v_j', v_k) = \delta_{jk}$. Then*

$$\Omega_B = \sum_j v_j \otimes v_j'$$

is invariant and independent of the basis.

Proof. The map $\phi\colon V \to V^*$, $\phi(w) = B(w, \cdot)$, is a linear isomorphism because B is nondegenerate. By the invariance of B and the dual action, ϕ is equivariant:

$$\phi(xw)(v) = B(xw, v) = -B(w, xv) = (x\phi(w))(v).$$

By construction $\phi(v_j') = v_j^*$, where (v_j^*) is the dual basis. Hence $\Omega_B = (\mathrm{id} \otimes \phi^{-1})\big(\sum_j v_j \otimes v_j^*\big)$. This is the image of the coevaluation tensor under an equivariant isomorphism. By Lecture 14, the coevaluation tensor is invariant and basis-independent. ∎

On $V = \mathbb{C}^N$ with $B(u, v) = u^{\mathsf{T}} v$, the Lie algebra $\mathfrak{so}_N$ consists of the matrices with $X^{\mathsf{T}} = -X$. The group $O_N = \{g : g^{\mathsf{T}} g = I\}$ acts on $V^{\otimes k}$ by $g^{\otimes k}$. The standard basis is orthonormal, so $e_i' = e_i$ and $\Omega = \sum_i e_i \otimes e_i$.

On $V = \mathbb{C}^{2n}$ with basis $e_1, \dots, e_n, f_1, \dots, f_n$, the symplectic form β is given by

$$\beta(e_i, f_j) = \delta_{ij} = -\beta(f_j, e_i), \qquad \beta(e_i, e_j) = \beta(f_i, f_j) = 0.$$

The partners of Lemma 27.6 are $e_i' = -f_i$ and $f_i' = e_i$. We use the sign-reversed invariant tensor

$$\Omega_{\mathrm{sp}} = -\Omega_\beta = \sum_{i=1}^{n} (e_i \otimes f_i - f_i \otimes e_i).$$

For even k and a perfect matching π of $\{1, \dots, k\}$ into pairs $a_j < b_j$, the *pair contraction* tensor Ω_π places the two factors of the jth copy of Ω in positions a_j, b_j. In other words, Ω_π comes from $\Omega^{\otimes k/2}$ by permuting tensor factors. Permuting factors is equivariant, so Ω_π is invariant. A *factor permutation* of a tensor in $V^{\otimes k} \otimes V^{*\otimes l}$ permutes the V-factors among themselves and the V^*-factors among themselves.

The *volume tensor* of $\mathbb{C}^N$ is

$$\mathrm{vol} = \sum_{\sigma \in S_N} \operatorname{sgn}(\sigma)\, e_{\sigma(1)} \otimes \cdots \otimes e_{\sigma(N)}.$$

By the Leibniz formula and the analogous calculation for $\mathfrak{gl}_N$,

$$g^{\otimes N}\mathrm{vol} = \det(g)\,\mathrm{vol}, \qquad X\,\mathrm{vol} = (\operatorname{tr} X)\,\mathrm{vol} \quad \text{for } g \in GL_N,\ X \in \mathfrak{gl}_N.$$

So vol is $\mathfrak{sl}_N$-invariant. Antisymmetric matrices have trace zero, so vol is also $\mathfrak{so}_N$-invariant. A reflection $r \in O_N$ has $\det r = -1$ and sends vol to $-$vol.

Theorem 27.7 (First fundamental theorems for tensors [C]). *Let V be the defining module and $k \geq 0$.*

(a) $(V^{\otimes k})^{O_N}$ *is zero for odd k. For even k it is spanned by the Ω_π.*

(b) For $N \geq 2$ (including the abelian case $N = 2$), $(V^{\otimes k})^{\mathfrak{so}_N}$ is the sum of $(V^{\otimes k})^{O_N}$ and the span of the factor permutations of $\mathrm{vol} \otimes \Omega^{\otimes (k-N)/2}$. *The second part occurs only if $k \geq N$ and $k - N$ is even.*

(c) $(V^{\otimes k})^{\mathfrak{sp}_{2n}}$ *is zero for odd k. For even k it is spanned by the symplectic Ω_π.*

(d) $(V^{\otimes k} \otimes V^{*\otimes l})^{\mathfrak{sl}_n} = 0$ *unless n divides $k - l$. First reorder tensor products into $V^{\otimes k} \otimes V^{*\otimes l}$. The invariants are spanned by factor permutations of products of pairings $\sum_i e_i \otimes e_i^*$ with $(k-l)/n$ copies of* vol *if $k \geq l$, or with $(l-k)/n$ copies of the dual volume tensor if $l \geq k$.*

For GL, O and Sp, see Goodman–Wallach, §§5.3.1–5.3.2. For the volume terms, see Fulton–Harris, Appendix F.1, Propositions F.10 and F.15. Polarisation gives the multilinear invariants. For SL_n, the dual volume identifies dual factors with exterior powers. We import these tensor forms. Passing to the Lie algebras of the connected groups SL_n, SO_N, Sp_{2n} preserves invariants: see Goodman–Wallach, §1.7.1, Proposition 1.7.7 and its proof.

The orthogonal volume term depends on the tensor degree. If $k < N$, there is no volume term, so $(V^{\otimes k})^{\mathfrak{so}_N}$ is spanned by pair contractions. We call $k < N$ the *stable range*. Linear independence of the Ω_π is a separate question. It fails when $\dim V$ is small. Exercise Sheet 27 shows this for the symplectic form on $\mathbb{C}^2$.

Since $V \cong V^*$ through B, Proposition 27.1 gives

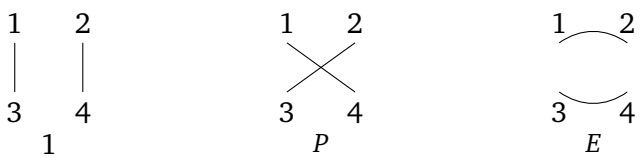
$$\operatorname{End}_{\mathfrak{g}}(V^{\otimes d}) \cong \left(V^{*\otimes d} \otimes V^{\otimes d}\right)^{\mathfrak{g}} \cong (V^{\otimes 2d})^{\mathfrak{g}}.$$

The relevant tensor degree is $2d$, not d. In the orthogonal case, for $d = 2$ the three matchings of four points give the operators

$$1, \qquad P(u \otimes v) = v \otimes u, \qquad E(u \otimes v) = B(u, v)\,\Omega.$$

In the symplectic case use $E(u \otimes v) = \beta(u, v)\Omega_{\mathrm{sp}}$.

The three orthogonal matchings are shown below. Positions $1, 2$ are outputs; positions $3, 4$ are inputs, identified with dual factors by B.



Each joining line pairs the indices at its endpoints.

Proposition 27.8 (Odd orthogonal commutants). *Let N be odd and $d \geq 1$. An operator on $V^{\otimes d}$ commutes with every $g^{\otimes d}$, $g \in SO_N$, if and only if it commutes with every $g^{\otimes d}$, $g \in O_N$.*

Proof. Since $SO_N \subseteq O_N$, one direction is clear. Conversely, let $g \in O_N$ with $\det g = -1$. Then

$$\det(-g) = (-1)^N \det g = 1,$$

so $h = -g \in SO_N$. Moreover, $g^{\otimes d} = (-1)^d h^{\otimes d}$, and a scalar multiple of $h^{\otimes d}$ commutes with everything that commutes with $h^{\otimes d}$. ∎

For even N and $2d \geq N$, the orthogonal invariant theorem gives further invariants. The tensor $\mathrm{vol} \otimes \Omega^{\otimes(2d-N)/2}$ is $\mathfrak{so}_N$-invariant, but a reflection negates it. Its operator lies in $\mathrm{End}_{\mathfrak{so}_N}(V^{\otimes d})$, not in the O_N-commutant. For $N = 4$ and $d = 2$ it is a nonzero multiple of the Hodge star on $\Lambda^2 V$, extended by zero on $\mathrm{Sym}^2 V$.

27.4 The representation ring

Write $\nu \leq \mu$ if $\mu - \nu$ is a sum of simple roots with nonnegative integer coefficients. For $\lambda \in P_+$, the *orbit sum* is

$$o_\lambda = \sum_{\mu \in W\lambda} e^\mu.$$

By Lecture 9, each W-orbit in P meets P_+ exactly once. Therefore the o_λ, $\lambda \in P_+$, form a $\mathbb{Z}$-basis of $\mathbb{Z}[P]^W$.

Definition 27.9 (Representation ring). The *representation ring* $K(\mathfrak{g})$ is the free abelian group with basis $[L(\lambda)]$, $\lambda \in P_+$. A module V has class

$$[V] = \sum_{\lambda \in P_+} \dim \mathrm{Hom}_{\mathfrak{g}}\big(L(\lambda), V\big)\, [L(\lambda)],$$

and multiplication extends $[L(\lambda)] \cdot [L(\mu)] = [L(\lambda) \otimes L(\mu)]$ bilinearly.

Complete reducibility and Lecture 14 give $[V] = [W]$ if and only if $V \cong W$, as well as $[V \oplus W] = [V] + [W]$. Since the tensor product distributes over direct sums, $[V][W] = [V \otimes W]$. Associativity, commutativity and the unit $[\mathbb{C}]$ follow from the module isomorphisms of Lecture 14.

By complete reducibility, every short exact sequence $0 \to U \to V \to W \to 0$ of modules splits. Hence $K(\mathfrak{g})$ agrees with the Grothendieck group defined by the relations $[V] = [U] + [W]$ for such sequences.

Lemma 27.10 (Unitriangular bases). *Let $S \subseteq P_+$ be finite and downward closed in P_+. Let $\mathbb{Z}_S$ be the span of $\{o_\mu : \mu \in S\}$. Suppose that for each $\mu \in S$,*

$$a_\mu = o_\mu + \sum_{\nu \in S,\ \nu < \mu} c_{\mu\nu} o_\nu, \qquad c_{\mu\nu} \in \mathbb{Z}.$$

Then $(a_\mu)_{\mu \in S}$ is a $\mathbb{Z}$-basis of $\mathbb{Z}_S$.

Proof. List S so that $\nu < \mu$ implies that ν comes first. The matrix of the a_μ in the basis o_μ is then unitriangular with integer entries. Its determinant is 1, so its inverse has integer entries. ∎

By Lecture 9, every finite set $F \subseteq P_+$ lies in the finite downward-closed set

$$\{\nu \in P_+ : \nu \le \lambda \text{ for some } \lambda \in F\}.$$

Theorem 27.11 (Character isomorphism)**.** *The map* $\operatorname{ch}\colon K(\mathfrak{g}) \to \mathbb{Z}[P]^W$, $[V] \mapsto \operatorname{ch} V$, *is a ring isomorphism.*

Proof. By Lecture 15, characters are additive, multiplicative and Weyl invariant, and $\operatorname{ch}\mathbb{C} = 1$. So ch is a ring homomorphism into $\mathbb{Z}[P]^W$.

The weights of $L(\lambda)$ are $\le \lambda$, and λ has multiplicity one (Lecture 12). Weyl invariance therefore gives

$$\operatorname{ch} L(\lambda) = o_\lambda + \sum_{\nu \in P_+,\ \nu<\lambda} \dim L(\lambda)_\nu \, o_\nu.$$

Take any $x \in K(\mathfrak{g})$ and $f \in \mathbb{Z}[P]^W$. Choose a finite, downward-closed set S containing the support of x in the simple-class basis and that of f in the orbit-sum basis. The characters $\operatorname{ch} L(\mu)$, $\mu \in S$, lie in $\mathbb{Z}_S$, because S is downward closed.

Lemma 27.10 shows that ch maps the span of $\{[L(\mu)] : \mu \in S\}$ bijectively onto $\mathbb{Z}_S$. Hence $\operatorname{ch} x = 0$ forces $x = 0$, and $f = \operatorname{ch} y$ for some y in the span of the $[L(\mu)]$, $\mu \in S$. ∎

Let $r = \operatorname{rank}\mathfrak{g}$ and $\chi_i = \operatorname{ch} L(\omega_i)$. For $\mu = \sum_i \mu_i \omega_i \in P_+$, put

$$\chi^\mu = \prod_{i=1}^{r} \chi_i^{\mu_i}.$$

Theorem 27.12 (Fundamental characters)**.** *The ring* $\mathbb{Z}[P]^W$ *is generated by* $\chi_1, \dots, \chi_r$, *which are algebraically independent. Hence* $K(\mathfrak{g}) \cong \mathbb{Z}[x_1, \dots, x_r]$, *with* $x_i \mapsto [L(\omega_i)]$.

Proof. The element χ^μ is the character of $T_\mu = \bigotimes_i L(\omega_i)^{\otimes \mu_i}$. A weight of T_μ is a sum $\nu = \sum_f \nu_f$ over the factors, with each ν_f at most the highest weight of its factor. Thus

$$\mu - \nu = \sum_f (\text{highest weight of } f - \nu_f) \ \ge 0.$$

Equality holds only if every ν_f is highest. Highest weight spaces are one-dimensional, so μ has multiplicity one. Since χ^μ is also Weyl invariant,

$$\chi^\mu = o_\mu + \sum_{\nu \in P_+,\ \nu<\mu} c_{\mu\nu} o_\nu.$$

For a finite, downward-closed S, Lemma 27.10 makes $(\chi^\mu)_{\mu\in S}$ a basis of $\mathbb{Z}_S$. Every element of $\mathbb{Z}[P]^W$ lies in some $\mathbb{Z}_S$, so the monomials χ^μ span. This proves generation.

A polynomial relation among the χ_i is a linear relation among monomials $\prod_i \chi_i^{a_i} = \chi^{\sum a_i \omega_i}$. Distinct exponent vectors give distinct μ, because the ω_i form a basis of P. Finitely many of these monomials lie in one basis $(\chi^\mu)_{\mu\in S}$, so they are linearly independent. Theorem 27.11 transfers the result to $K(\mathfrak{g})$. ∎

For example, for $\mathfrak{sl}_2$ let $x = [V_1]$. Clebsch–Gordan gives

$$x[V_m] = [V_{m+1}] + [V_{m-1}] \quad \text{for } m \ge 1.$$

The cases $m = 1$ and $m = 2$ give

$$[V_2] = x^2 - 1, \qquad [V_3] = x[V_2] - [V_1] = x^3 - 2x.$$

27.5 Worked example: the vector square

Let $N \geq 5$, $\mathfrak{g} = \mathfrak{so}_N$ and $V = \mathbb{C}^N$. By Lecture 23,

$$V \otimes V = \mathbb{C}\Omega \oplus \Lambda^2 V \oplus \operatorname{Sym}^2_0 V,$$

where $\operatorname{Sym}^2_0 V$ is the kernel of the linear map induced by B on $\operatorname{Sym}^2 V$. The summands are irreducible for $N \geq 5$. Their dimensions

$$1, \qquad \frac{N(N-1)}{2}, \qquad \frac{N(N+1)}{2} - 1$$

are pairwise distinct, so the summands are pairwise nonisomorphic.

Degree two. $(V \otimes V)^{\mathfrak{g}} = \mathbb{C}\Omega$.

Degree three. Via B and Proposition 27.1, $(V^{\otimes 3})^{\mathfrak{g}} \cong \operatorname{Hom}_{\mathfrak{g}}(V, V \otimes V)$. The three dimensions above differ from N when $N \geq 5$, so Schur's lemma makes this space zero. This agrees with the stable range, since $3 < N$.

Degree four. Theorem 27.4 gives $\operatorname{End}_{\mathfrak{g}}(V \otimes V) \cong \mathbb{C}^3$. This matches the three matchings of four points, since $4 < N$ lies in the stable range. Write $B(\Omega)$ for the linear map induced by B applied to Ω. The operators $1, P, E$ satisfy

$$B(\Omega) = \sum_i B(e_i, e_i) = N, \qquad E^2 = NE, \qquad PE = EP = E.$$

Here $E^2 = NE$ because $B(\Omega) = N$. Moreover, $PE = E$ because $P\Omega = \Omega$, and $EP = E$ because B is symmetric. The orthogonal idempotents are

$$p_0 = \frac{1}{N}E, \qquad p_- = \frac{1}{2}(1 - P), \qquad p_+ = \frac{1}{2}(1 + P) - \frac{1}{N}E.$$

Their images are $\mathbb{C}\Omega$, $\Lambda^2 V$ and $\operatorname{Sym}^2_0 V$. Under the isomorphism Ψ of Theorem 27.4, they correspond to the identity operators on the three one-dimensional multiplicity spaces.

The same method applies to the vector squares of $\mathfrak{sl}_n$ and $\mathfrak{sp}_{2n}$ from Lecture 18, Lecture 21 and Lecture 22. Exercise 27.6 constructs the symplectic projectors and compares the three cases.

Self-checks

(1) What is $\dim \operatorname{End}_{\mathfrak{sl}_2}(V_1^{\otimes 3})$?

(2) For $N \geq 2$, when is $(V^{\otimes 3})^{\mathfrak{so}_N} \neq 0$?

(3) Express $[V_4]$ in $K(\mathfrak{sl}_2) = \mathbb{Z}[x]$.

Answers. (1) $V_1^{\otimes 3} \cong V_3 \oplus 2V_1$, so the dimension is $1^2 + 2^2 = 5$. (2) Only for $N = 3$. Odd degree has no O_N part, and the volume part needs $3 \geq N$ with $3 - N$ even. (3) $[V_4] = x[V_3] - [V_2] = x^4 - 3x^2 + 1$. The dimensions check: $16 - 12 + 1 = 5$.

Summary

Intertwiners are invariant tensors. Multiplicity spaces determine commutants. Classical invariant tensors are built from pairings and volume tensors.

Characters identify $K(\mathfrak{g})$ with $\mathbb{Z}[P]^W$, a polynomial ring in the fundamental characters. Lecture 28 turns to the centre of $U(\mathfrak{g})$.

Reading

Fulton–Harris, Appendix F.1, and Goodman–Wallach, §5.3, give the invariant theorems cited above. Kirillov, §8.6, treats multiplicities. Exercise Sheet 27 gives further calculations.